

ERNEST WONG

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EDUCATION

Simon Fraser University | BSc, Computing Science Expected December 2027
University of Glasgow | Field Studies in Cybercrime August 2024
British Columbia Institute of Technology | Cloud Computing Systems Diploma, with Distinction December 2021

TECHNICAL SKILLS

Languages: Python, Go, Java, C#, TypeScript, JavaScript, SQL, Bash
Systems: React, Node.js, GraphQL, REST, Linux, AWS, Docker, Kubernetes, OpenShift, Salesforce
Data and Delivery: Pandas, scikit-learn, Denodo, Power BI, Jenkins, GitHub Actions, Selenium, JUnit, Pytest, Git

EXPERIENCE

Oakmont Industries Ltd. September 2026 - Present
Software Engineering Co-op Student | AI Systems Integration Surrey, BC

- Turned an open-ended ERP integration request into a vendor-backed proof of concept for moving eMullion sales orders into Aquilon without re-keying order data.
- Validated Aquilon's XML and JSON import path over SFTP, mapped eMullion fields to its sales-order schema, and produced a testable Go implementation plan.

Elections Canada May 2026 - Present
AI Engineer | Data Scientist Co-op Remote

- Replaced hand transcription with a Python pipeline that converts multi-page Tableau PDFs into normalized CSVs for Power BI and recovers metrics missed in earlier reports.
- Made the pipeline resilient to new reporting years by splitting extraction into bilingual parsers for five dashboard sections and validating decimals, missing fields, and the distinction between zero and null.
- Moved repeated Power BI rules into reusable Denodo views, letting non-technical analysts filter flagged records and calculate issue rates without rewriting SQL.

Innovation, Science and Economic Development Canada January 2025 - August 2026
Site Reliability Engineer | Cloud Operations | Student Programmer Co-op Ottawa, ON

- Delivered maintenance-window releases for a \$100M+ patent submission platform by coordinating AWS, Salesforce, Java, database, testing, and validation work across teams.
- Replaced manual environment-by-environment diagnosis with a Java and GraphQL drift detector that compares 8+ environments daily in Jenkins and points engineers to configuration mismatches.
- Standardized release setup for 50+ applications by automating Bitbucket, Bamboo, Nexus, and IBM WebSphere configuration with Python and Selenium.

SFU Satellite Design Team | UBC Orbit September 2023 - September 2026
Executive | Mission Operations Developer | Embedded Systems Lead Burnaby, BC

- Took a high-altitude balloon mission from design reviews through flight at approximately 30 km, integrating Python flight software with 10+ sensors and subsystems on a Raspberry Pi.
- Built ALEASAT's React mission-operations dashboard for telemetry monitoring and command workflows, supporting preparations for a European Space Agency mission and planned 2028 SpaceX launch.

PROJECTS

CleanListen | Python, scikit-learn 2025 - Present

- Reduced the text a screen reader must narrate by an estimated 48-52% while preserving research content, using a classifier that reached 91.2% accuracy and 89.7% KEEP F1 across 1,929 labeled PDF lines.
- Packaged the model as an installable CLI with reproducible training, cleaning, and benchmarking, backed by Pytest, GitHub Actions on Python 3.10 and 3.12, and document-grouped evaluation splits.

ShareGuard | JavaScript, Chrome MV3, Playwright 2026

- Built a Chrome Manifest V3 extension that masks passwords, API tokens, and selected regions in a separate local audience view while leaving the host page unchanged, with navigation recovery and Chromium smoke tests.

AI Cybersecurity Research | Simon Fraser University May 2024 - August 2024

- Exceeded 80% F1 on personal-data removal with a Python privacy filter, then used Tensor Trust prompt-injection attacks to expose sensitive-data leakage the classifier still missed.

LEADERSHIP

- Hosted ISED AI Coffee Chats for 300+ attendees and presented SFU Computing Science to 100+ prospective students.

AWARDS AND CERTIFICATIONS

AI Hackathon Finalist: Most Impactful Project and Best Use of Google Gemini
SFU Computing Science Undergraduate Society Award: 1 of 5 recipients annually
Global Skills Opportunity Award: EDI International Mobility
Mayor of Richmond Recognition Award: Community contributions
Certifications: AWS Certified AI Practitioner | Amateur Radio Operator Certificate (VE7 OOH)